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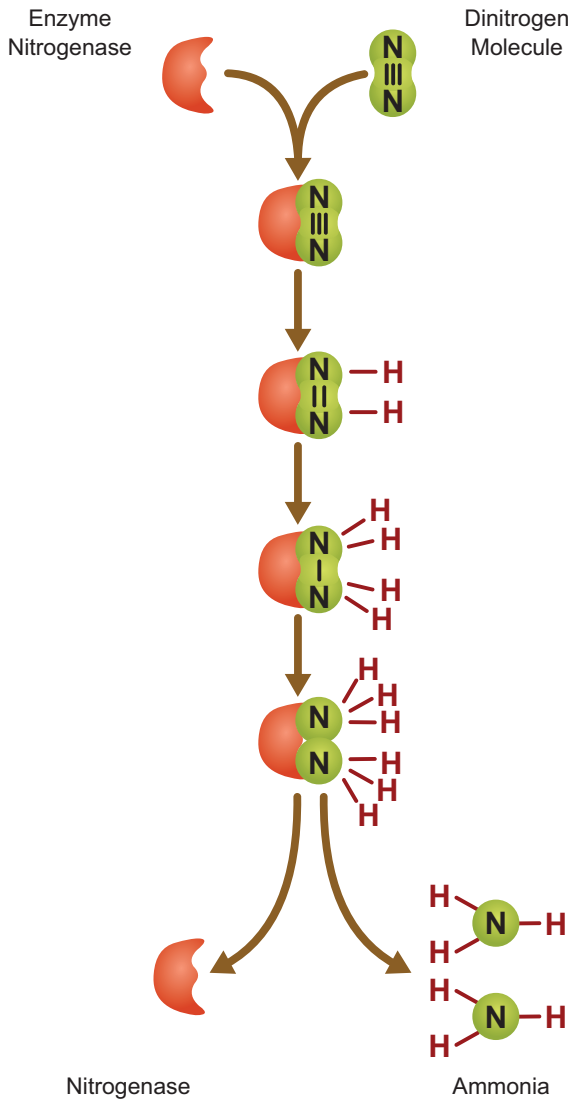


Figure 12.7: Nitrogenase enzyme function

Overall equation:

$$N_2 + 8e^- + 8H^+ + 16ATP \longrightarrow 2NH_3^+ + H_2 + 16ADP + 16\ Pi$$

2. Nitrification

Ammonia (NH_3^+) is converted into Nitrite (NO_2^-) by *Nitrosomonas* bacterium. Nitrite is then converted into Nitrate (NO_3^-) by *Nitrobacter* bacterium.

Plants are more adapted to absorb nitrate (NO_3^-) than ammonium ions from the soil.

$$2\ NH_3^+ + 3\ O_2 \xrightarrow{Nitrosomonas} 2\ NO_2^- + 2\ H^+ + 2H_2O$$

$$2\ NO_2^- + O_2 \xrightarrow{Nitrobacter} 2\ NO_3^-$$

3. Nitrate Assimilation

The process by which nitrate is reduced to ammonia is called **nitrate assimilation** and occurs during nitrogen cycle.

$$NO_3^- \xrightarrow[Mo]{\text{Nitrate reductase}} NO_2^-$$

$$NO_2^- \xrightarrow[Cu,\ Fe]{\text{Nitrite reductase}} NH_3^+$$

4. Ammonification

Decomposition of organic nitrogen (proteins and amino acids) from dead plants and animals into ammonia is called **ammonification**. Organisim involved in this process are *Bacillus ramosus* and *Bacillus vulgaris*.

5. Denitrification

Nitrates in the soil are converted back into atmospheric nitrogen by a process called **denitrification**. Bacteria involved in this process are *Pseudomonas*, *Thiobacillus* and *Bacillus subtilis*.

$$\text{Nitrate } (NO_3^-) \xrightarrow{Pseudomonas} \text{Molecular Nitrogen } (N_2)$$

The overall process of nitrogen cycle is given in Figure 12.8.

12.8.2 Nitrogen Metabolism

Ammonium Assimilation (Fate of Ammonia)

Ammonia is converted into amino acids by the following processes:

1. Reductive amination

Glutamic acid or glutamate is formed by reaction of ammonia with α-ketoglutaric acid.